## PEER REVIEW HISTORY

BMJ Open publishes all reviews undertaken for accepted manuscripts. Reviewers are asked to complete a checklist review form (**http://bmjopen.bmj.com/site/about/resources/checklist.pdf**) and are provided with free text boxes to elaborate on their assessment. These free text comments are reproduced below.

## ARTICLE DETAILS

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| **TITLE (PROVISIONAL)** | Statistical Methods for Non-adherence in Non-inferiority Trials: Useful and Used? A Systematic Review |
| **AUTHORS** | Dodd, Matthew; Fielding, Katherine; Carpenter, James; Thompson, Jennifer; Elbourne, Diana |

## VERSION 1 – REVIEW

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| **REVIEWER** | Bell, Melanie<br>University of Arizona, Epidemiology and Biostatistics |
| **REVIEW RETURNED** | 15-Jun-2021 |

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| **GENERAL COMMENTS** | Review of Statistical Methods for Non-adherence in Non-inferiority Trials: Useful and Used? A Systematic Review<br><br>Major comment<br>1. The major limitation of this review, which the manuscript does a nice job of discussing, is the need for the term adherence to be included in the title, abstract or keywords. This represents a potentially serious selection bias, and makes the review much less useful, and perhaps misleading, as it makes it seem like most trials account for non-adherence. I realize that I am suggesting a lot of extra work, but the authors could do a random sample of NI trials, and report on how they handled non-adherence.<br>2. Including methodological papers in a review on what trialists are using to handle non-adherence doesn't make sense to me, and doesn't seem to align with the objectives of the paper, as stated in the last paragraph of the introduction.<br>3. Examples of non-adherence, including the connection between non-adherence and missing data, ought to be briefly outlined.<br>4. I found the explanation of Instrumental Variables to not be very clear. While Tables 3 and 4 give more detail, some ought to be given in the text. While I am only passingly familiar with these methods, a reader with my understanding or less might wonder, for example, what the instrument is for trials (as IV methods are usually used for non-randomized studies).<br>5. Related to the previous comment, Tables 3 and 4 are so technical that it is likely that only statisticians with familiarity of each of the methods would be able to follow them.<br><br>Minor comments<br>1. I'm not sure that I agree with the statement "… mITT analyses are not normally used to account for the impact of non-adherence." Some researchers may include non-adherence in their definition of mITT. |

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| | 2. In Table 1, the journal doesn't really add much and makes the table quite long. Perhaps add to supplementary file instead?<br>3. Why is power included in Table 1?<br>4. Why are there sometimes 4 simulations studies and other times 3, in Table 1?<br>5. In Table 3 the first column has the label "Method (estimand)". I don't believe that the word "estimand" should be there, as I think most statisticians working in trials think about estimands with the ICH-E9R1 definition, which has 5 components to it (Population, Treatment condition, Variable, Inter current events and how to handle, Summary measure for the variable). |

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| **REVIEWER** | Gillespie, David<br>Centre for Trials Research, University of Cardiff |
| **REVIEW RETURNED** | 18-Oct-2021 |

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| **GENERAL COMMENTS** | General<br>I like the idea of this paper and I think it is well written.<br><br>My overarching concern relates to the search strategy. The authors exclude papers which did not contain terms related to adherence and statistical methods for handling non-adherence in the titles, abstracts and key words. Can the authors explain why they did this? I presume that this was viewed as a sensible and likely efficient approach for identifying particular statistical methods used in non-inferiority trials to adjust for non-adherence. However, by limiting the scope in this way I'm not sure how the authors can meaningfully "quantify the use of such methods".<br><br>Ultimately, my comments on this are minor – this is a useful systematic review of statistical methods and should be helpful for statisticians working on non-inferiority trials.<br>I outline section-specific comments below in their respective sections.<br><br>Methods<br><br>Only one database searched. Would the authors consider this as a limitation? If so, this should be mentioned in the Discussion and the potential implications of focussing on one database only.<br><br>The authors do not describe their synthesis approach.<br><br>Results<br>The advantages and disadvantages are not discussed in the results. If they are an outcome I think it is worth explicitly mentioning that no RCTs described them. I see that they are contained in Table 3. The authors should edit the text in the manuscript to link this more clearly.<br><br>Discussion<br>I wonder what the authors could comment on the use of IV methods and the estimation of the complier average causal effect in particular when treatment is not all or nothing (i.e. could be several doses of medicine or several sessions of a psychological intervention). The paper puts forward a strong case for CACE, but with these sorts of interventions I believe there are some strong |

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| | assumptions required about the exclusion restriction (depending on how “adherence” is defined in this particular instance) that may not be ameliorated with effective double blinding.<br><br>Furthermore, there are plenty of interventions which cannot be evaluated in a double-blind setting. Would the authors consider amending the statement on Page 12 to “In individually randomised trials...are typically satisfied by effective double blinding and/or use of objective outcomes” or would you argue that the use of objective outcomes does not overcome compensate for a lack of blinding in this particular situation?<br>The authors go on to discuss the administering of interventions at multiple time points. However, I would recommend that these sections are weaved together a bit more to highlight what a trial statistician may need to consider when weighing up a CACE approach over IPTW.<br><br>Page 14 – authors describe the limited number of statistical analysis plans provided following requests. I think this should be reported in the results. |

**VERSION 1 – AUTHOR RESPONSE**

Comment 8 (Reviewer 1)

The major limitation of this review, which the manuscript does a nice job of discussing, is the need for the term adherence to be included in the title, abstract or keywords. This represents a potentially serious selection bias, and makes the review much less useful, and perhaps misleading, as it makes it seem like most trials account for non-adherence. I realize that I am suggesting a lot of extra work, but the authors could do a random sample of NI trials, and report on how they handled non-adherence.

Thanks for your comments. We think that the reviewer may have meant to say that "…it makes it seem like most trials did not account for non-adherence". Within the papers that reached full-text review, several mentioned non-adherence in some context (for instance, when describing the non-adherence observed within a particular trial or when defining the per-protocol population as a subgroup of the wider intention-to-treat population) but very few actually applied a statistical method in order to account for non-adherence to interventions. The introduction section of the review has been edited in order to clarify that our main aim was to identify such methods:

"This systematic review aimed to identify statistical methods that can be used to account for the impact of non-adherence to interventions (thereby estimating the causal effects of experimental interventions) in randomised non-inferiority trials. Secondary aims were to quantify the use of such methods in these studies and examine their impact on non-inferiority conclusions."

Following the reviewer's suggestion, we have also reviewed a random sample of 50 non-inferiority trials (protocols or results papers). The Ovid MEDLINE database was searched (using all fields) for terms related to "randomised" and "non-inferior" within papers published up to 31st December 2020.

Of the 50 trials reviewed, none included statistical methods of interest, 26 performed/planned to perform “per-protocol” analyses and one performed an “on-treatment” analysis.

Comment 9 (Reviewer 1)

Including methodological papers in a review on what trialists are using to handle non-adherence doesn’t make sense to me, and doesn’t seem to align with the objectives of the paper, as stated in the last paragraph of the introduction.

Thanks for raising this point. Given that our main aim was to identify statistical methods that can be used to account for non-adherence to interventions in randomised non-inferiority trials, we feel that including methodological papers that have applied such methods within the context of non-inferiority trials is appropriate. In other words, we intended to include both methods that have been applied within non-inferiority trials and also methods which are suitable for use in these studies but may not have been applied yet (e.g. the tipping point approach). The introduction section has been edited in order to clarify that this was our main aim:

"This systematic review aimed to identify statistical methods that can be used to account for the impact of non-adherence to interventions (thereby estimating the causal effects of experimental interventions) in randomised non-inferiority trials. Secondary aims were to quantify the use of such methods in these studies and examine their impact on non-inferiority conclusions."

Comment 10 (Reviewer 1)

Examples of non-adherence, including the connection between non-adherence and missing data, ought to be briefly outlined.

Thanks for your suggestions. The following text has now been added to the introduction section:

"Examples of non-adherence include not receiving a surgical intervention as planned, not taking all of the prescribed doses of a medication, or not attending all of the sessions of an exercise rehabilitation program.”

“Non-adherence may also be linked with missing outcome data if, for example, the trial protocol stipulates that further follow-up is no longer required once adherence drops below a specific threshold or if non-adherent participants become lost to follow-up."

Comment 11 (Reviewer 1)

I found the explanation of Instrumental Variables to not be very clear. While Tables 3 and 4 give more detail, some ought to be given in the text. While I am only passingly familiar with these methods, a reader with my understanding or less might wonder, for example, what the instrument is for trials (as IV methods are usually used for non-randomized studies).

Apologies that the explanation of IV methods was not clear in the main text. The following text has now been added in the discussion:

"IV methods use randomisation as the instrument in order to account for unmeasured confounders of the outcome and intervention received (i.e. adherence)."

"When these assumptions hold, it is relatively straightforward to show that if we regress the intervention received (i.e. adherence) on randomisation, and then use this model to predict each participant's adherence, these predictions are orthogonal (independent) of all adherence-outcome confounders. Therefore, if in a second step we regress the outcome on these predictions, we get an unconfounded estimate of the effect of adherence on outcome. It follows that, in contrast to techniques that involve inverse-probability weighting, when the above four IV assumptions hold, IV methods enable us to estimate CACE even in the presence of unmeasured confounding (although inclusion of measured confounders can improve precision).[32, 33] While IV methods may thus appear a panacea, as usual in statistics, there are no free lunches: a lack of precision and statistical power is often a challenge with IV techniques and methods used to adjust for non-adherence more generally.[5, 30, 34, 35]"

Comment 12 (Reviewer 1)

Related to the previous comment, Tables 3 and 4 are so technical that it is likely that only statisticians with familiarity of each of the methods would be able to follow them.

Thanks for raising this important point. We agree that some aspects of the tables are quite technical and therefore have added an additional table in the appendix (Table A2) which describes the formulae presented in Tables 3 and 4 in more detail. It is worth noting that statisticians working on non-inferiority trials would form a substantial proportion of our target audience and that references to papers describing the methods in further detail have also been included in Tables 3 and 4.

Comment 13 (Reviewer 1)

I'm not sure that I agree with the statement "… mITT analyses are not normally used to account for the impact of non-adherence." Some researchers may include non-adherence in their definition of mITT.

Thanks for raising this point. The sentence quoted from our review and the comment from the reviewer appear to be consistent with each other. We also recognise that the modified intention-to-treat population is not consistently defined.

Comment 14 (Reviewer 1)

In Table 1, the journal doesn't really add much and makes the table quite long. Perhaps add to supplementary file instead?

Thanks for this suggestion. The data on journal has now been moved to the supplementary material.

Comment 15 (Reviewer 1)

Why is power included in Table 1?

Data on the power of each trial was included in case this was informative when considering the results looking at the impact of the statistical methods on non-inferiority conclusions. However, we agree with the reviewer that this information is not essential and it has now been removed in order to shorten Table 1.

Comment 16 (Reviewer 1)

Why are there sometimes 4 simulations studies and other times 3, in Table 1?

Thanks for highlighting this. One of the simulation studies included in the review was based on a real trial and therefore the unit of randomisation for this analysis was previously categorised as 'individual'. This has now been re-categorised as 'simulation study' in order to show that there were four simulation studies included in total.

Comment 17 (Reviewer 1)

In Table 3 the first column has the label “Method (estimand)”. I don’t believe that the word “estimand” should be there, as I think most statisticians working in trials think about estimands with the ICH-E9R1 definition, which has 5 components to it (Population, Treatment condition, Variable, Inter current events and how to handle, Summary measure for the variable).

Thanks for raising this issue. As the reviewer mentions, the ICH-E9R1 definition uses "estimand" as an umbrella term containing five components (population, treatment, variable of interest, intercurrent event handling, and summary measure for the variable). In this review, we are focussed on the intercurrent event component of the estimand. Rather than removing the term "estimand" completely, we have clarified that we are focussing on the intercurrent event component of the definition in the footnote to Table 3.

Comment 18 (Reviewer 2)

I like the idea of this paper and I think it is well written.

My overarching concern relates to the search strategy. The authors exclude papers which did not contain terms related to adherence and statistical methods for handling non-adherence in the titles, abstracts and key words. Can the authors explain why they did this? I presume that this was viewed as a sensible and likely efficient approach for identifying particular statistical methods used in non-inferiority trials to adjust for non-adherence. However, by limiting the scope in this way I’m not sure how the authors can meaningfully “quantify the use of such methods”.

Ultimately, my comments on this are minor – this is a useful systematic review of statistical methods and should be helpful for statisticians working on non-inferiority trials.

I outline section-specific comments below in their respective sections.

Thanks for your encouraging remarks. We hypothesised that the proportion of eligible papers would be low and therefore this search strategy was used in order to identify statistical methods that can be used to account for non-adherence in non-inferiority trials in an efficient way. Quantifying the use of the methods was not our main aim and so the introduction section has been edited in order to clarify this:

"This systematic review aimed to identify statistical methods that can be used to account for the impact of non-adherence to interventions (thereby estimating the causal effects of experimental interventions) in randomised non-inferiority trials. Secondary aims were to quantify the use of such methods in these studies and examine their impact on non-inferiority conclusions."

Whilst there are limitations to our search strategy (which are clearly acknowledged in the discussion), this review is able to provide some idea regarding how often statistical methods for handling non-adherence have been used in the context of non-inferiority trials. In response to comment number 8, we also reviewed a random sample of 50 non-inferiority trials (protocols or results papers) and found that none used a statistical method of interest.

Comment 19 (Reviewer 2)

Methods: Only one database searched. Would the authors consider this as a limitation? If so, this should be mentioned in the Discussion and the potential implications of focussing on one database only.

Thanks for highlighting that this limitation was omitted and we agree that it should be included. Please see our response to comment number 1.

Comment 20 (Reviewer 2)

Methods: The authors do not describe their synthesis approach.

Thanks for raising this point. The following text has now been added in the methods section:

“Information was largely combined using a narrative synthesis approach i.e. “synthesis of findings from multiple studies that relies primarily on the use of words and text to summarise and explain the findings of the synthesis”.[27]”

Comment 21 (Reviewer 2)

Results: The advantages and disadvantages are not discussed in the results. If they are an outcome I think it is worth explicitly mentioning that no RCTs described them. I see that they are contained in Table 3. The authors should edit the text in the manuscript to link this more clearly.

Thanks for raising this important point. The following text has now been added to the results section:

"Advantages and disadvantages of the statistical methods

The advantages and disadvantages of the methods identified (as stated by the authors) are shown in Table 3. Advantages or disadvantages of the techniques used were stated in eight (33%) of the 24 papers included; six were methodological papers and two were results papers. No advantages or disadvantages were stated for five of the 11 methods identified."

Comment 22 (Reviewer 2)

Discussion: I wonder what the authors could comment on the use of IV methods and the estimation of the complier average causal effect in particular when treatment is not all or nothing (i.e. could be several doses of medicine or several sessions of a psychological intervention). The paper puts forward a strong case for CACE, but with these sorts of interventions I believe there are some strong assumptions required about the exclusion restriction (depending on how “adherence” is defined in this particular instance) that may not be ameliorated with effective double blinding.

Thanks for your suggestion. The following text has now been added to the discussion:

"The two-stage least-squares regression (2SLS) approach sketched in the previous paragraph can be applied when the intervention is not all or nothing. Suppose that a non-inferiority trial is conducted to assess whether prescribing one dose of a medication per week is non-inferior to prescribing two doses per week over the course of four weeks. For each participant, the monotonicity assumption requires that the potential number of doses taken would be lower if the participant was randomly assigned to receive one dose per week than if they were randomised to receive two doses per week. Assuming there are no covariates and the monotonicity assumption holds, it can be shown that the 2SLS estimator converges towards a weighted average of the causal effects of one unit increases in the intervention amongst compliers (individuals whose intervention intensity is affected by randomisation (the instrument)).[36, 37] This is because the implicit effect of the 2SLS analysis is that values of the outcome at which there are more compliers get given greater weight."

Comment 23 (Reviewer 2)

Discussion: Furthermore, there are plenty of interventions which cannot be evaluated in a double-blind setting. Would the authors consider amending the statement on Page 12 to “In individually randomised trials...are typically satisfied by effective double blinding and/or use of objective outcomes” or would you argue that the use of objective outcomes does not overcome compensate for a lack of blinding in this particular situation?

Thanks for this suggestion. We are happy to amend the sentence as you suggest:

"In individually randomised trials, the exclusion restriction and monotonicity assumptions are typically satisfied by effective double-blinding and/or use of objective outcomes, and the exchangeability assumption is usually valid since randomisation is expected to produce trial arms that are balanced with respect to prognostic factors."

Comment 24 (Reviewer 2)

Discussion: The authors go on to discuss the administering of interventions at multiple time points. However, I would recommend that these sections are weaved together a bit more to highlight what a trial statistician may need to consider when weighing up a CACE approach over IPTW.

Thanks for this suggestion. The following text has now been added to the discussion:

"When considering whether to apply an IV approach or a G-method, statisticians might consider whether the exclusion restriction and monotonicity assumptions are realistic given the context of the trial, and whether randomisation is a sufficiently strong instrument. Where outcomes are collected at multiple timepoints, inverse-probability weighting may be a more attractive approach if data on potential confounders are also collected throughout follow-up."

Comment 25 (Reviewer 2)

Discussion: Page 14 – authors describe the limited number of statistical analysis plans provided following requests. I think this should be reported in the results.

Thanks for this suggestion. The following text has now been added to the results section:

"Statistical analysis plans
Statistical analysis plans were requested for all 17 non-inferiority trials where the protocol or results paper was included in the review, and obtained for nine of these trials."

**VERSION 2 – REVIEW**

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| **REVIEWER** | Bell, Melanie<br>University of Arizona, Epidemiology and Biostatistics |
| **REVIEW RETURNED** | 15-Dec-2021 |

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| **GENERAL COMMENTS** | All of my concerns have been addressed, thank you for the opportunity to review this interesting paper. |

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| **REVIEWER** | Gillespie, David<br>Centre for Trials Research, University of Cardiff |
| **REVIEW RETURNED** | 16-Dec-2021 |

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| **GENERAL COMMENTS** | Thank you for responding to my feedback points. I am happy with the edits made and recommend this article for publication. |